Habib University

EE 354 / CE 361 / MATH 310 – Introduction to Probability and Statistics

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Assignment No. 02

Release Date: [28/09/2026]

Due by: [11/10/2026]

Total points: 120

Points obtained:

Student Name:	Student ID:	Section:

EE 354 / CE 361 / MATH 310 – Introduction to Probability and Statistics
Assignment No. 02

Purpose:
The purpose of this assignment is to help you practice writing and working with probability mass functions, finding the distribution of a function of a random variable, computing expectation and variance, working with joint distributions of two or more random variables, and reasoning about conditional probability and independence.

Instructions:

1. This assignment must be completed individually.
2. All answers must be written in dark ink.
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Assignments will be graded by the instructor/TA.
2. You must show a reasonable amount of work. Marks may be deducted if important steps are skipped over or missing entirely.
3. You may be called for a viva to demonstrate your understanding. If you cannot correctly answer the question you have attempted right, you may lose your marks.
4. A grade of zero will be awarded if plagiarism is detected.
5. Untidy work may result in a deduction of points.

Late Submission Penalty:

- 1-day late submission - 20% deduction of the maximum allowable marks
- No submission will be accepted after one day of the original deadline

CLO Assessment:
This assignment assesses the following course learning outcomes:

Course Learning Outcomes	**CLO Assessed**
CLO 1: Transform a description of a random experiment into the formal setting of probability theory – construct sample space, identify and specify events. (Cog-4)	
CLO 2: Apply the axioms of probability and compute probabilities by counting. (Cog-3)	
CLO 3: Compute probabilities of events and moments from given distributions. (Cog-3)	✓
CLO 4: Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog-3)	✓
CLO 5: Analyze a random experiment for presence of dependent situations and modeling and analyzing it using notions of conditional probability. (Cog-4)	

Total Marks: 120

Problem Statement 1 [10 Points] - Cash-back

A store's spinner promotion has 10 equally likely outcomes, split among 5 distinct cash-back rewards as follows: 1 sector awards \$0.20, 2 sectors award \$0.40, 2 sectors award \$0.50, 3 sectors award \$0.80, and 2 sectors award \$1.00. A customer spins once; let X denote the cash-back reward received, in dollars.

(a) State the two defining properties that any valid PMF P_X must satisfy. [2]

(b) Find the range R_X of X (i.e., the set of all values that X can take), and construct the PMF $P_X(x)$ from the description of the spinner. Verify that your PMF satisfies the properties from part (a). [2]

(c) Find $P(X \leq 0.5)$. [2]

(d) Find $P(0.25 < X < 0.75)$. [2]

(e) Find $P(X = 0.2 \mid X < 0.6)$. [2]

Problem Statement 2 [10 Points] - Escape Room

You are locked in an escape room with a countdown timer. You have already identified the one drawer that contains the clue you need, but the key that opens it is not marked. You are holding a ring of 5 identical-looking keys, exactly one of which opens this specific drawer, and you begin trying them, one at a time, on the drawer's lock. Let X denote the number of trials you need to open it.

(a) Suppose that once a key fails to open the drawer, you set it aside and never try it again, so that every subsequent trial is made from the keys not yet tried. Find the PMF $P_X(x)$ of X under this assumption. [3]

(b) Suppose instead that you are too rattled to keep track of which keys you have already tried, so at every trial you pick uniformly at random from all 5 keys, possibly retrying a key that has already failed. Find the PMF $P_X(x)$ of X under this assumption. [3]

(c) Without doing any additional calculation, explain in one or two sentences why X has a finite range under the assumption in part (a), but can in principle take arbitrarily large values under the assumption in part (b). [2]

(d) Suppose the timer only leaves you enough time for 2 attempts. Using your answers to parts (a) and (b), compute $P(X \leq 2)$ under each assumption, and state which assumption gives you better odds of opening the drawer in time. [2]

Problem Statement 3 [10 Points] - Common Birthday

You go to a party with 500 guests, and you would like to know the chance that someone else at the party shares your birthday. Assume that each guest's birthday is equally likely to be any of the 365 days of the year, that all guests' birthdays are independent of one another, and that (for simplicity) no one is born on February 29. Let X denote the number of *other* guests (out of the 500) whose birthday falls on the same day as yours.

(a) Find the range R_X of X. Then, using the assumptions given above, identify the probability p that a single given guest shares your birthday. [2]

(b) Using your answer to part (a), derive the PMF $P_X(x)$ of X from first principles (i.e., by directly counting/reasoning about which guests could share your birthday, rather than quoting a formula). Use your PMF to compute $P(X = 1)$ exactly, i.e., the probability that exactly one other guest shares your birthday. [3]

(c) State the conditions under which a Poisson PMF is commonly used to approximate a Binomial PMF, and briefly explain why they are satisfied here. Compute the parameter λ of the approximating Poisson PMF, and use it to approximate $P(X = 1)$. Compare your approximate answer to the exact answer from part (b). [3]

(d) The classical "birthday paradox" asks for the probability that, among a group of only 20–30 people, *some* two people share a birthday, and this probability turns out to be surprisingly high (over 50% for just 23 people). In contrast, this problem asks for the probability that someone shares *your* specific birthday among 500 people, and the answer turns out to be fairly small. In one or two sentences, explain why these two probabilities behave so differently, even though 500 is much larger than 23. [2]

Problem Statement 4 [10 Points] - Functions of Random Variables

Let X be a discrete random variable with the following PMF:

$$p_X(k) = \begin{cases} \frac{1}{4} & k = -2 \\ \frac{1}{8} & k = -1 \\ \frac{1}{8} & k = 0 \\ \frac{1}{4} & k = 1 \\ \frac{1}{4} & k = 2 \\ 0 & \text{otherwise} \end{cases}$$

Define a new random variable $Y = (X + 1)^2$.

(a) Find the range R_Y of Y. [4]

(b) Find the PMF $p_Y(y)$ of Y. [6]

Problem Statement 5 [10 Points] - Expectation & Variance

The discrete random variable X has the probability mass function

$$p_X(x) = \begin{cases} kx & x = 2, 4, 6 \\ k(x-2) & x = 8, 9, 10 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of k. [2]

(b) What is $P(X = 5)$? [2]

(c) Find $E[X]$ and $E[X^2]$. [2]

(d) Find $\text{Var}(X)$. [2]

(e) Define a new random variable $Y = 3 - 5X$. Find $E[Y]$ and $\text{Var}(Y)$. [2]

Problem Statement 6 [10 Points] - Two Rolls of a Dice

Consider the following random variables X and Y:

X = the magnitude of the difference between the outcomes of two rolls of a fair 3-sided die. (For example, if the first roll results in 1 and the second roll results in 3, the value taken by X is 2. Similarly, if the first roll results in 3 and the second roll results in 1, the value taken by X is 2.)

$Y = 2X$.

(a) Calculate and plot the PMF of X. [2]

(b) Calculate and plot the PMF of Y. [2]

(c) Calculate the mean of Y. [3]

(d) Calculate the variance of Y. [3]

Problem Statement 7 [10 Points] - Monsoon Season

A Habib University student lives in a hostel that is 2 miles from campus. If the weather is clear, which happens with probability 0.6, he walks the entire way to campus at a speed of 5 miles per hour. However, during Karachi's monsoon season, the streets near campus often flood, so whenever the weather is bad (with probability 0.4), he instead rides his motorbike the entire way at a speed of 30 miles per hour. Let V denote his speed (in miles per hour) on a given day, and let T denote the time (in hours) it takes him to get to campus on that day. Recall that time is related to speed and distance by

$$T = \frac{\text{distance}}{V} = \frac{2}{V}.$$

(a) Find the PMF $p_T(t)$ of T. [2]

(b) Using the PMF from part (a), compute $E[T]$, the mean time it takes him to get to campus. [3]

(c) Compute $E[V]$, the mean of his speed. [3]

(d) A classmate argues that the mean travel time can also be found by simply computing $\frac{2}{E[V]}$ (distance divided by mean speed). Compute this quantity, compare it to your answer in part (b), and explain in one or two sentences why this approach does not give the correct mean travel time in general. [2]

Problem Statement 8 [10 Points] - Joint PMFs of Multiple Random Variables

Question 1. The random variables X and Y have the joint PMF $p_{X,Y}(x, y)$ given in the table below, where a, b, and c are unknown constants.

	$x = 2$	$x = 4$	$x = 6$
$y = 6$	$\frac{1}{12}$	$\frac{1}{12}$	b
$y = 4$	$\frac{2}{12}$	a	$\frac{2}{12}$
$y = 2$	$\frac{1}{12}$	$\frac{1}{12}$	c

It is also known that Y is equally likely to take each of its three possible values, i.e., $p_Y(2) = p_Y(4) = p_Y(6) = \frac{1}{3}$.

(a) Using the fact that a valid PMF must sum to 1, together with the given information about $p_Y(y)$, determine the values of a, b, and c. [2]

(b) Find the marginal PMF $p_X(x)$. [2]

(c) Compute $P(X > Y)$. [2]

(d) Define a new random variable $Z = X + Y$. Recall that for a function $g(X, Y)$ of two random variables, the expected value rule states

$$E[g(X, Y)] = \sum_x \sum_y g(x, y)\, p_{X,Y}(x, y).$$

Using this formula with $g(x, y) = x+y$, compute $E[Z]$ directly from the joint PMF $p_{X,Y}(x, y)$. Then, using the marginal PMF $p_X(x)$ from part (b) and the given marginal PMF of Y, compute $E[X]$ and $E[Y]$ separately, and verify that $E[X] + E[Y]$ matches your answer for $E[Z]$. [4]

Problem Statement 9 [10 Points] - Linear Algebra

At Habib University, a large Linear Algebra class has 300 students, and each student independently has a probability of $\frac{1}{3}$ of failing the course. For each student i (where $i = 1, 2, \ldots, 300$), define

$$X_i = \begin{cases} 1 & \text{if student } i \text{ fails the course} \\ 0 & \text{if student } i \text{ passes the course.} \end{cases}$$

Let $X = X_1 + X_2 + \cdots + X_{300}$ denote the total number of students who fail the course.

(a) Write down the PMF of X_i, and use it to compute $E[X_i]$. [4]

(b) Recall that linearity of expectation states

$$E[X_1 + X_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n],$$

and that this holds regardless of whether the random variables involved are independent. Using this fact and your answer to part (a), find $E[X]$, the expected number of students who fail the course. [4]

(c) Suppose that due to a change in course policy, only 250 of the 300 students are now at risk of failing (the remaining 50 are graded on a separate curve and cannot fail), and the failure probability for each of these 250 students rises to $\frac{2}{5}$. Using the same approach as in parts (a) and (b), find the new expected number of students who fail the course. [2]

Problem Statement 10 [10 Points] - Conditioning

Consider two random variables X and Y with the joint PMF $p_{X,Y}(x, y)$ given in the table below.

	$Y = 1$	$Y = 2$
$X = 1$	$\frac{1}{3}$	$\frac{1}{12}$
$X = 2$	$\frac{1}{6}$	0
$X = 4$	$\frac{1}{12}$	$\frac{1}{3}$

(a) Find the marginal PMFs $p_X(x)$ and $p_Y(y)$. [3]

(b) Find $P(X \leq 2,\ Y > 1)$. [2]

(c) Find the conditional PMF $p_{Y|X}(y \mid 1)$, and use it to state $P(Y = 2 \mid X = 1)$. [2]

(d) Now suppose instead that we are told $Y = 1$. Find the conditional PMF $p_{X|Y}(x \mid 1)$, and use it to compute $E[X \mid Y = 1]$ and $\mathrm{Var}(X \mid Y = 1)$. [3]

Problem Statement 11 [10 Points] - What's your CGPA?

At Habib University, 40% of students belong to the School of Arts, Humanities and Social Sciences (AHSS), and the remaining 60% belong to the School of Science and Engineering (DSSE). Let X be the random variable indicating whether a randomly selected HU student belongs to DSSE, i.e.,

$$X = \begin{cases} 1 & \text{if the student belongs to DSSE} \\ 0 & \text{if the student belongs to AHSS.} \end{cases}$$

Let Y denote the CGPA of a randomly selected HU student. You are given the following information:

$$E[Y \mid X = 1] = 3.0, \qquad E[Y \mid X = 0] = 3.2.$$

Suppose you stand at the main entrance gate of HU, ask the CGPA of the first 100 students who walk in, and calculate their average CGPA.

(a) Write down the PMF of X. [3]

(b) State the total expectation theorem for a random variable Y conditioned on a random variable X, and briefly explain in words what each term in the formula represents. [3]

(c) Using the theorem from part (b), the PMF from part (a), and the given conditional expectations, compute your best guess for the average CGPA that would be obtained from this exercise. Briefly justify why this approach is correct, rather than simply averaging 3.0 and 3.2 directly. [4]

Problem Statement 12 [10 Points] - Independence

Consider the following joint PMF $p_{X,Y}(x, y)$ of two random variables X and Y:

	$x = 0$	$x = 1$	$x = 2$	$x = 3$
$y = 4$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$y = 2$	0	$\frac{1}{8}$	$\frac{1}{8}$	0
$y = 1$	0	0	$\frac{1}{8}$	$\frac{1}{8}$

(a) Are X and Y independent? Justify your answer by checking the definition of independence (i.e., find a pair of values (x, y) for which $p_{X,Y}(x, y) \neq p_X(x)\, p_Y(y)$, if one exists). [2]

(b) Find $p_{X|Y}(x \mid 1)$ and $p_{Y|X}(y \mid 1)$. [2]

(c) Find $E[X \mid Y = 4]$. [2]

(d) Let $R = \max(X, Y)$. Find $E[R]$. [2]

(e) Let $S = X + Y$. Find $E[S]$ and $\mathrm{Var}(S)$ directly from the PMF of S. Then check whether $E[S] = E[X] + E[Y]$ holds, and whether $\mathrm{Var}(S) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ holds. Briefly explain any difference in your findings for the two checks. [2]